

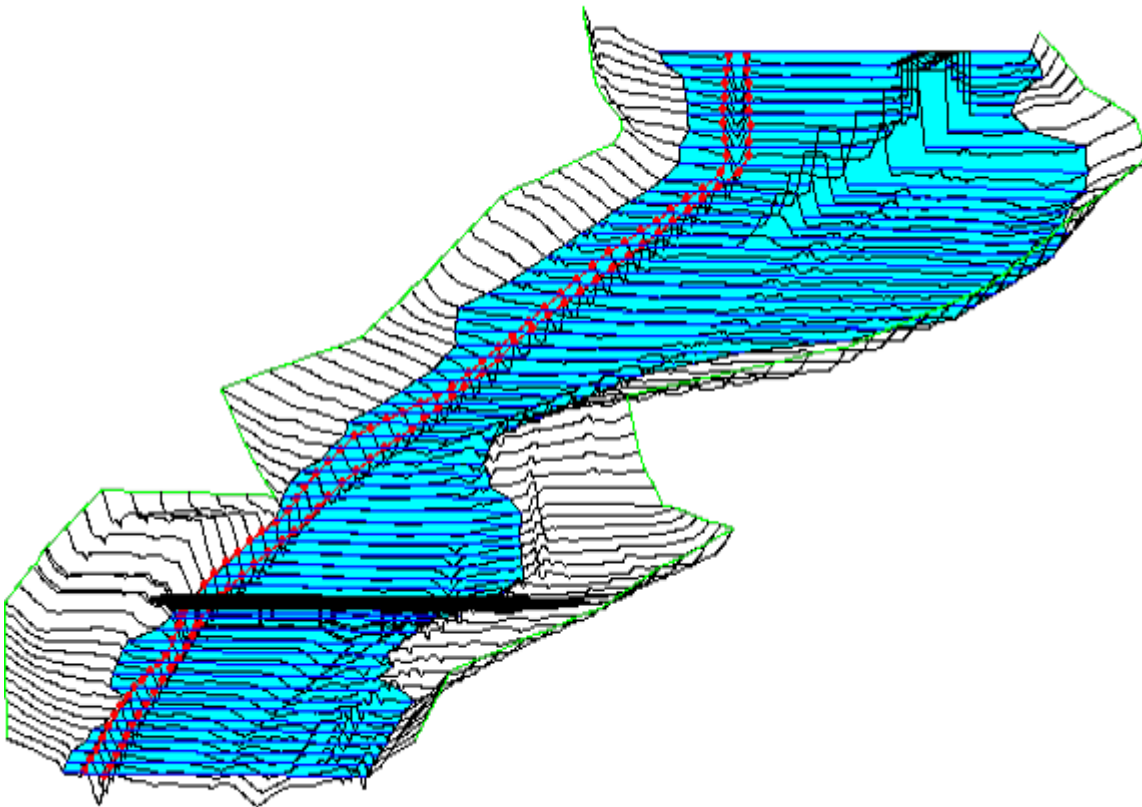


**US Army Corps
of Engineers®**

Hydrologic Engineering Center

HEC-RAS

River Analysis System



Culvert Hydraulics

Version 7.0

Exported - April 2026

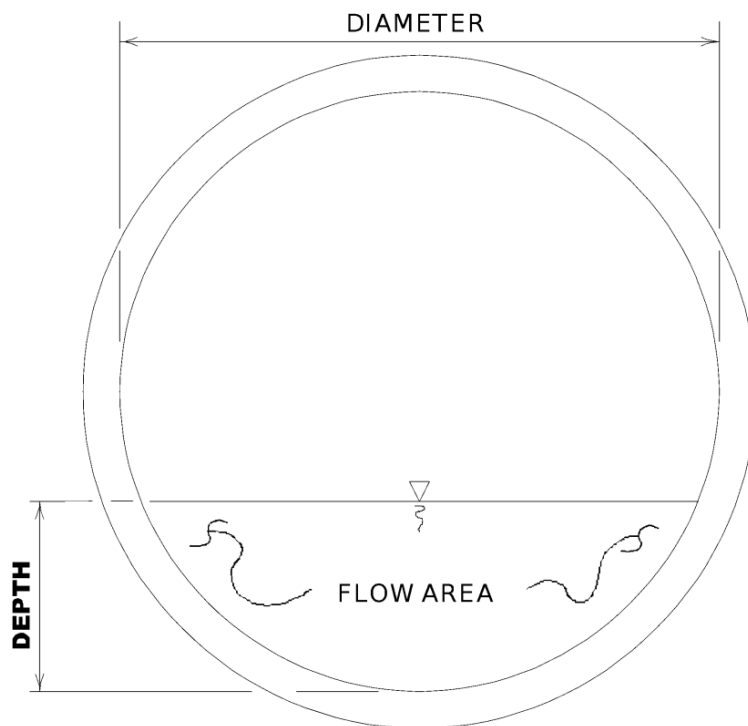
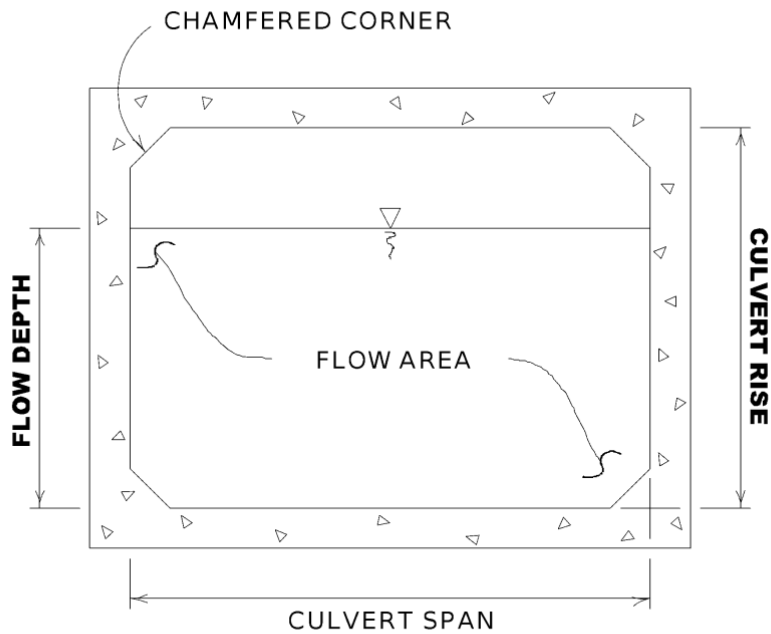
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This section introduces the basic concepts of culvert hydraulics, which are used in the HECRAS culvert routines.

1 INTRODUCTION TO CULVERT TERMINOLOGY

A culvert is a relatively short length of closed conduit, which connects two open channel segments or bodies of water. Two of the most common types of culverts are: circular pipe culverts, which are circular in cross section, and box culverts, which are rectangular in cross section. The figure below shows an illustration of circular pipe and box culverts. In addition to box and pipe culverts, HEC-RAS has the ability to model arch; pipe arch; low profile arch; high profile arch; elliptical; semi-circular; and ConSpan culvert shapes.



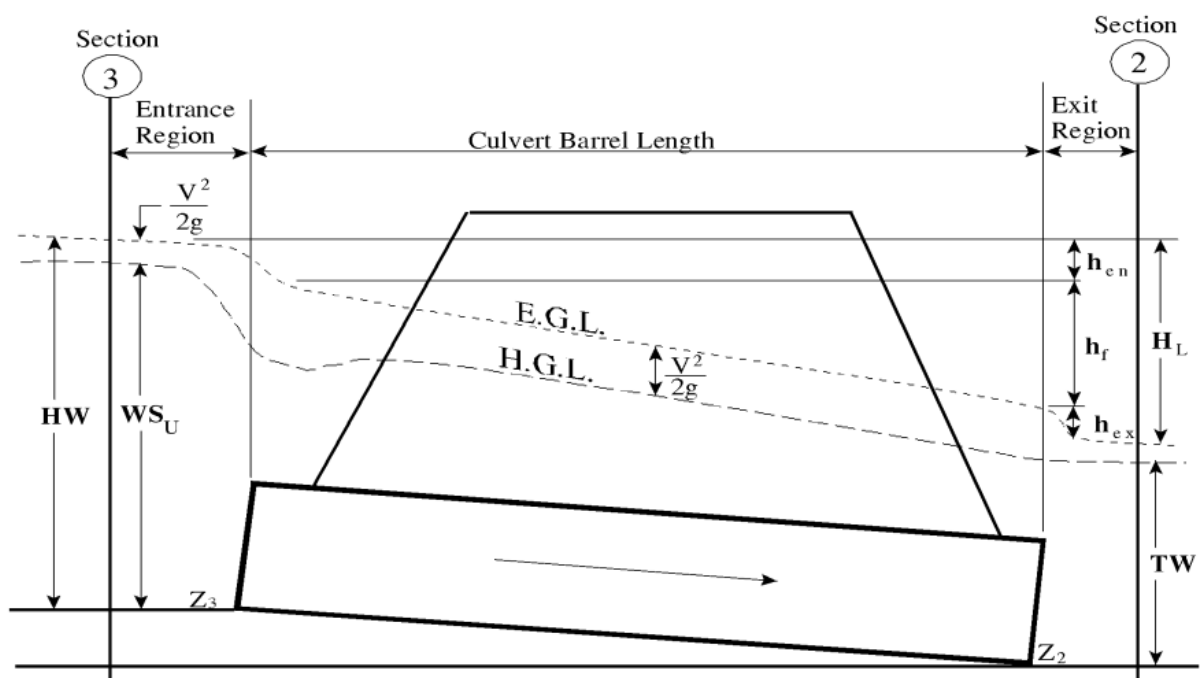
Culverts are made up of an **entrance** where water flows into the culvert, a **barrel**, which is the closed conduit portion of the culvert, and an **exit**, where the water flows out of the culvert (see

the figure below). The total flow capacity of a culvert depends upon the characteristics of the entrance as well as the culvert barrel and exit. The **Tailwater** at a culvert is the depth of water on the exit or downstream side of the culvert, as measured from the downstream invert of the culvert (shown as **TW** on the figure below). The **invert** is the lowest point on the inside of the culvert at a particular cross section. The tailwater depth depends on the flow rate and hydraulic conditions downstream of the culvert.

Headwater (HW on the figure below) is the depth from the culvert inlet invert to the energy grade line, for the cross section just upstream of the culvert (Section 3). The Headwater represents the amount of energy head required to pass a given flow through the culvert.

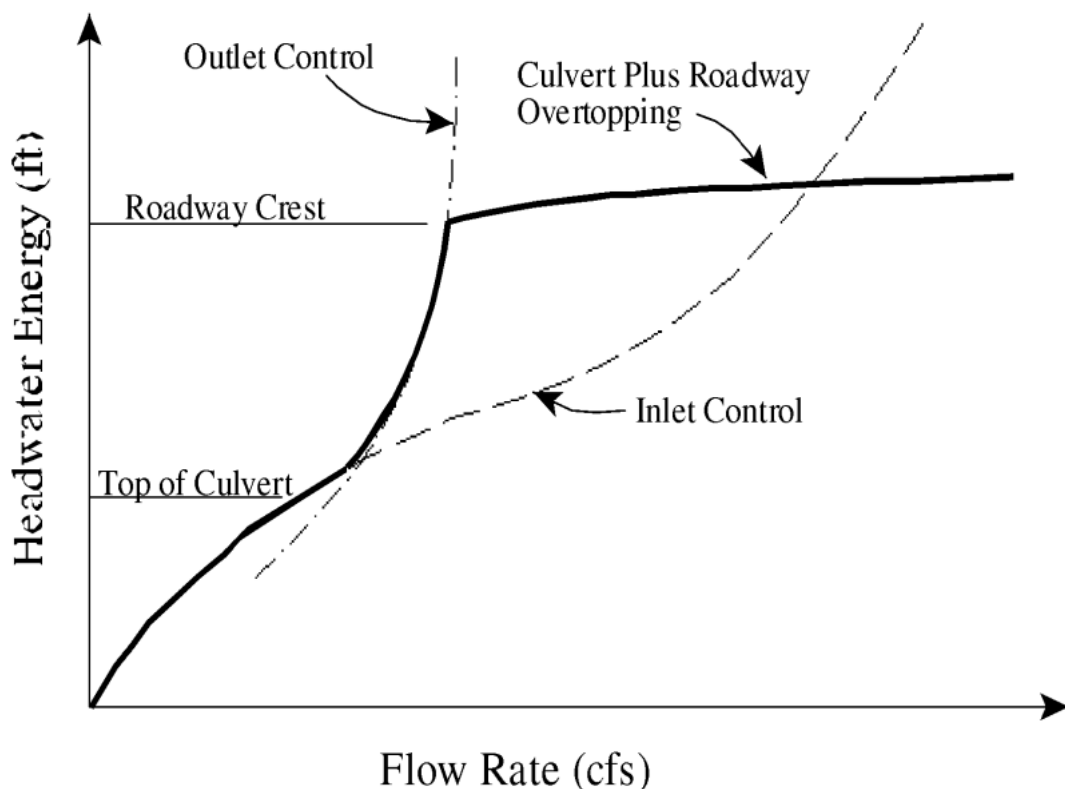
The **Upstream Water Surface** (WSU on the figure below) is the depth of water on the entrance or upstream side of the culvert (Section 3), as measured from the upstream invert of Cross Section 3.

The **Total Energy** at any location is equal to the elevation of the invert plus the specific energy (depth of water + velocity heady) at that location. All of the culvert computations within HEC-RAS compute the total energy for the upstream end of the culvert. The upstream water surface (WSU) is then obtained by placing that energy into the upstream cross section and computing the water surface that corresponds to that energy for the given flow rate.



2 FLOW ANALYSIS FOR CULVERTS

The analysis of flow in culverts is quite complicated. It is common to use the concepts of "inlet control" and "outlet control" to simplify the analysis. **Inlet control** flow occurs when the flow capacity of the culvert entrance is less than the flow capacity of the culvert barrel. The control section of a culvert operating under inlet control is located just inside the entrance of the culvert. The water surface passes through critical depth at or near this location, and the flow regime immediately downstream is supercritical. For inlet control, the required upstream energy is computed by assuming that the culvert inlet acts as a sluice gate or as a weir. Therefore, the inlet control capacity depends primarily on the geometry of the culvert entrance. **Outlet control** flow occurs when the culvert flow capacity is limited by downstream conditions (high tailwater) or by the flow carrying capacity of the culvert barrel. The HECRAS culvert routines compute the upstream energy required to produce a given flow rate through the culvert for inlet control conditions and for outlet control conditions (see figure below). In general, the higher upstream energy "controls" and determines the type of flow in the culvert for a given flow rate and tailwater condition (however, this is not always true). For outlet control, the required upstream energy is computed by performing an energy balance from the downstream section to the upstream section. The HECRAS culvert routines consider entrance losses, friction losses in the culvert barrel, and exit losses at the outlet in computing the outlet control headwater of the culvert.



During the computations, if the inlet control answer comes out higher than the outlet control answer, the program will perform some additional computations to evaluate if the inlet control answer can actually persist through the culvert without pressurizing the culvert barrel. The assumption of inlet control is that the flow passes through critical depth near the culvert inlet and transitions into supercritical flow. If the flow persists as low flow through the length of the culvert barrel, then inlet control is assumed to be valid. If the flow goes through a hydraulic jump inside the barrel, and fully develops the entire area of the culvert, it is assumed that this condition will cause the pipe to pressurize over the entire length of the culvert barrel and thus act more like an orifice type of flow. If this occurs, then the outlet control answer (under the assumption of a full flowing barrel) is used instead of the inlet control answer.

3 COMPUTING INLET CONTROL HEADWATER

For inlet control conditions, the capacity of the culvert is limited by the capacity of the culvert opening, rather than by conditions farther downstream. Extensive laboratory tests by the National Bureau of Standards, the Bureau of Public Roads, and other entities resulted in a series of equations, which describe the inlet control headwater under various conditions. These equations form the basis of the FHWA inlet control nomographs shown in the “Hydraulic Design of Highway Culverts” publication [FHWA, 1985]. The FHWA inlet control equations are used by the HEC-RAS culvert routines in computing the upstream energy. The inlet control equations were developed for submerged and unsubmerged inlet conditions. These equations are:

Unsubmerged Inlet:

$$1) \quad \frac{HW_i}{D} = \frac{H_c}{D} + K \left[\frac{Q}{AD^{0.5}} \right]^M - 0.5S$$

$$2) \quad \frac{HW_i}{D} = K \left[\frac{Q}{AD^{0.5}} \right]^M$$

Submerged Inlet:

$$3) \quad \frac{HW_i}{D} = c \left[\frac{Q}{AD^{0.5}} \right]^2 + Y = 0.5S$$

Symbol	Description	Units
HW_i	Headwater energy depth above the invert of the culvert inlet, feet	
D	Interior height of the culvert barrel	feet
H_c	Specific head at critical depth ($dc + V_c^2/2g$)	feet
Q	Discharge through the culvert	cfs
A	Full cross sectional area of the culvert barrel	feet ²
S	Culvert barrel slope	feet/feet

K, M, c, Y Equation constants, which vary depending on culvert shape and entrance conditions

Note that there are two forms of the unsubmerged inlet equation. The first form (1) is more correct from a theoretical standpoint, but form two (2) is easier to apply and is the only

documented form of equation for some of the culvert types. Both forms of the equations are used in the HEC-RAS software, depending on the type of culvert.

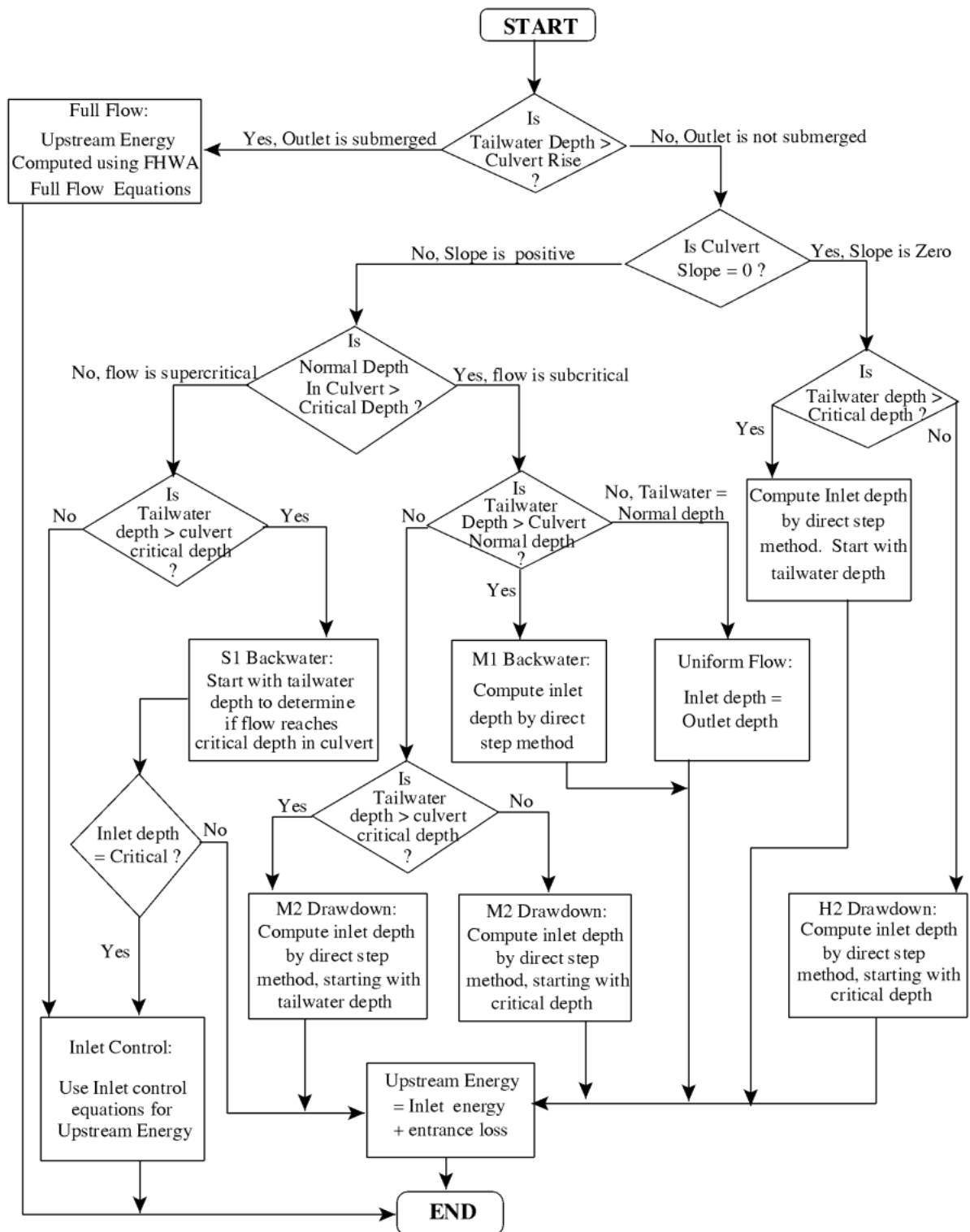
The nomographs in the FHWA report are considered to be accurate to within about 10 percent in determining the required inlet control headwater [FHWA, 1985]. The nomographs were computed assuming a culvert slope of 0.02 feet per foot (2 percent). For different culvert slopes, the nomographs are less accurate because inlet control headwater changes with slope. However, the culvert routines in HEC-RAS consider the slope in computing the inlet control energy. Therefore, the culvert routines in HEC-RAS should be more accurate than the nomographs, especially for slopes other than 0.02 feet per foot.

4 COMPUTING OUTLET CONTROL HEADWATER

For outlet control flow, the required upstream energy to pass the given flow must be computed considering several conditions within the culvert and downstream of the culvert. The figure below illustrates the logic of the outlet control computations. HEC-RAS use's Bernoulli's equation in order to compute the change in energy through the culvert under outlet control conditions. The outlet control computations are energy based. The equation used by the program is the following:

$$4) \quad Z_3 + Y_3 + \frac{a_3 V_3^2}{2g} = Z_2 + Y_2 + \frac{a_2 V_2^2}{2g} + H_L$$

Symbols	Description	Units
Z_3	Upstream invert elevation of the culvert	
Y_3	The depth of water above the upstream culvert inlet	
V_3	The average velocity upstream of the culvert	
a_3	The velocity weighting coefficient upstream of the culvert	
g	The acceleration of gravity	
Z_2	Downstream invert elevation of the culvert	
Y_2	The depth of water above the downstream culvert inlet	
V_2	The average velocity downstream of the culvert	
a_2	The velocity weighting coefficient downstream of the culvert	
H_L	Total energy loss through the culvert (from section 2 to 3)	



5 FHWA FULL FLOW EQUATIONS

For culverts flowing full, the total head loss, or energy loss, through the culvert is measured in feet (or meters). The head loss, H_L , is computed using the following formula:

$$5) \quad H_L = h_{en} + h_f + h_{ex}$$

Symbol	Description	Units
h_{en}	entrance loss	feet or meters
h_f	friction loss	feet or meters
h_{ex}	exit loss	feet or more

The friction loss in the culvert is computed using Manning's formula, which is expressed as follows:

$$6) \quad h_f = L \left(\frac{Qn}{1.486AR^{2/3}} \right)^2$$

Symbol	Description	Units
h_f	friction loss	feet
L	culvert length	feet
Q	flow rate in the culvert	cfs
n	Manning's roughness coefficient	
A	area of flow	square feet
R	hydraulic radius	feet

The exit energy loss is computed as a coefficient times the change in velocity head from just inside the culvert, at the downstream end, to outside of the culvert at the downstream end. The entrance loss is computed as a coefficient times the absolute velocity head of the flow inside the culvert at the upstream end. The exit and entrance loss coefficients are described in the next section of this chapter.

6 DIRECT STEP WATER SURFACE PROFILE COMPUTATIONS

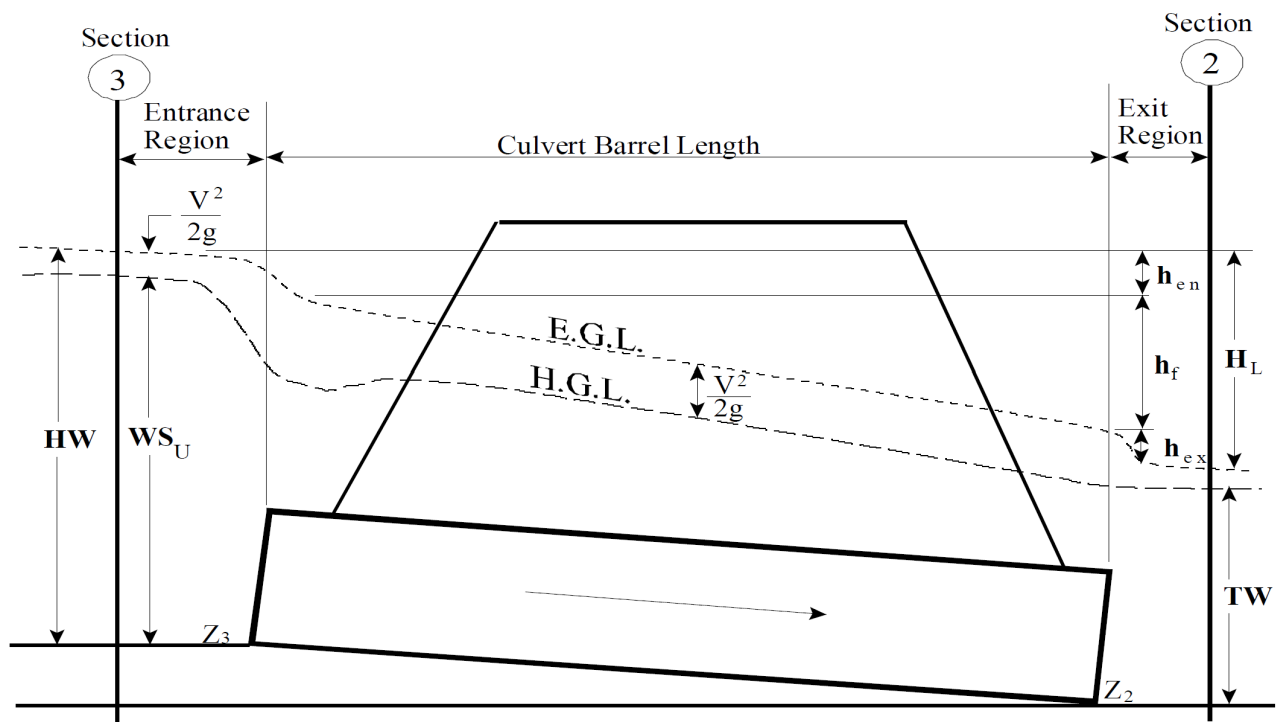
For culverts flowing partially full, the water surface profile in the culvert is computed using the direct step method. This method is very efficient, because no iterations are required to determine the flow depth for each step. The water surface profile is computed for small increments of depth (usually between 0.01 and 0.05 feet). If the flow depth equals the height of the culvert before the profile reaches the upstream end of the culvert, the friction loss through the remainder of the culvert is computed assuming full flow.

The first step in the direct step method is to compute the exit loss and establish a starting water surface inside the culvert. If the tailwater depth is below critical depth inside the culvert, then the starting condition inside the culvert is assumed to be critical depth. If the tailwater depth is greater than critical depth in the culvert, then an energy balance is performed from the downstream cross section to inside of the culvert. This energy balance evaluates the change in energy by the following equation.

$$7) \quad Z_c + Y_c + \frac{a_c V_c^2}{2g} = Z_2 + Y_2 + \frac{a_2 V_2^2}{2g} + H_{ex}$$

Symbol	Description	Units
Z_c	Elevation of the culvert invert at the downstream end	
Y_c	Depth of flow inside culvert at downstream end	
V_c	Velocity inside the culvert at downstream end	
Z_2	Invert elevation of the cross section downstream of culvert (Cross Section 2 from the figure below)	
Y_2	Depth of water at Cross Section 2	
V_2	Average velocity of flow at Section 2	

Once a water surface is computed inside the culvert at the downstream end, the next step is to perform the direct step backwater calculations through the culvert. The direct step backwater calculations will continue until a water surface and energy are obtained inside the culvert at the upstream end. The final step is to add an entrance loss to the computed energy to obtain the upstream energy outside of the culvert at Section 3 (see figure below). The water surface outside the culvert is then obtained by computing the water surface at Section 3 that corresponds to the calculated energy for the given flow rate.



Full flowing culvert with energy and hydraulic grade lines

7 NORMAL DEPTH OF FLOW IN THE CULVERT

Normal depth is the depth at which uniform flow will occur in an open channel. In other words, for a uniform channel of infinite length, carrying a constant flow rate, flow in the channel would be at a constant depth at all points along the channel, and this would be the normal depth.

Normal depth often represents a good approximation of the actual depth of flow within a channel segment. The program computes normal depth using an iterative approach to arrive at a value, which satisfies Manning's equation:

$$8) \quad Q = \frac{1.486}{n} A R^{2/3} S_f^{1/2}$$

Symbol	Description	Units
Q	flow rate in the channel	cfs
n	Manning's roughness coefficient	
A	area of flow	square feet
R	hydraulic radius	feet
S_f	slope of energy grade line	feet per foot

If the normal depth is greater than the culvert rise (from invert to top of the culvert), the program sets the normal depth equal to the culvert rise.

8 CRITICAL DEPTH OF FLOW IN THE CULVERT

Critical depth occurs when the flow in a channel has a minimum specific energy. **Specific energy** refers to the sum of the depth of flow and the velocity head. Critical depth depends on the channel shape and flow rate.

The depth of flow at the culvert outlet is assumed to be equal to critical depth for culverts operating under outlet control with low tailwater. Critical depth may also influence the inlet control headwater for unsubmerged conditions.

The culvert routines compute critical depth in the culvert by an iterative procedure, which arrives at a value satisfying the following equation:

$$9) \quad \frac{Q^2}{g} = \frac{A^3}{T}$$

Symbol	Description	Units
Q	flow rate in the channel	cfs
g	acceleration due to gravity, 32.2	ft/sec ²
A	cross-sectional area of flow	square feet
T	Top width of flow	feet

Critical depth for box culverts can be solved directly with the following equation [AISI, 1980]:

$$10) \quad y_c = \sqrt[3]{\frac{q^2}{g}}$$

Symbol	Description	Units
y_c	critical depth	feet
q	unit discharge per linear foot of width	cfs/ft
g	acceleration due to gravity, 32.2	ft/sec ²

9 HORIZONTAL AND ADVERSE CULVERT SLOPES

The culvert routines also allow for horizontal and adverse culvert slopes. The primary difference is that normal depth is not computed for a horizontal or adverse culvert. Outlet control is either computed by the direct step method for an unsubmerged outlet or the full flow equation for a submerged outlet.

10 WEIR FLOW

The first solution through the culvert is under the assumption that all of the flow is going through the culvert barrels. Once a final upstream energy is obtained, the program checks to see if the energy elevation is greater than the minimum elevation for weir flow to occur. If the computed energy is less than the minimum elevation for weir flow, then the solution is final. If the computed energy is greater than the minimum elevation for weir flow, the program performs an iterative procedure to determine the amount of flow over the weir and through the culverts. During this iterative procedure, the program recalculates both inlet and outlet control culvert solutions for each estimate of the culvert flow. In general the higher of the two is used for the culvert portion of the solution, unless the program feels that inlet control cannot be maintained. The program will continue to iterate until it finds a flow split that produces the same upstream energy (within the error tolerance) for both weir and culvert flow.

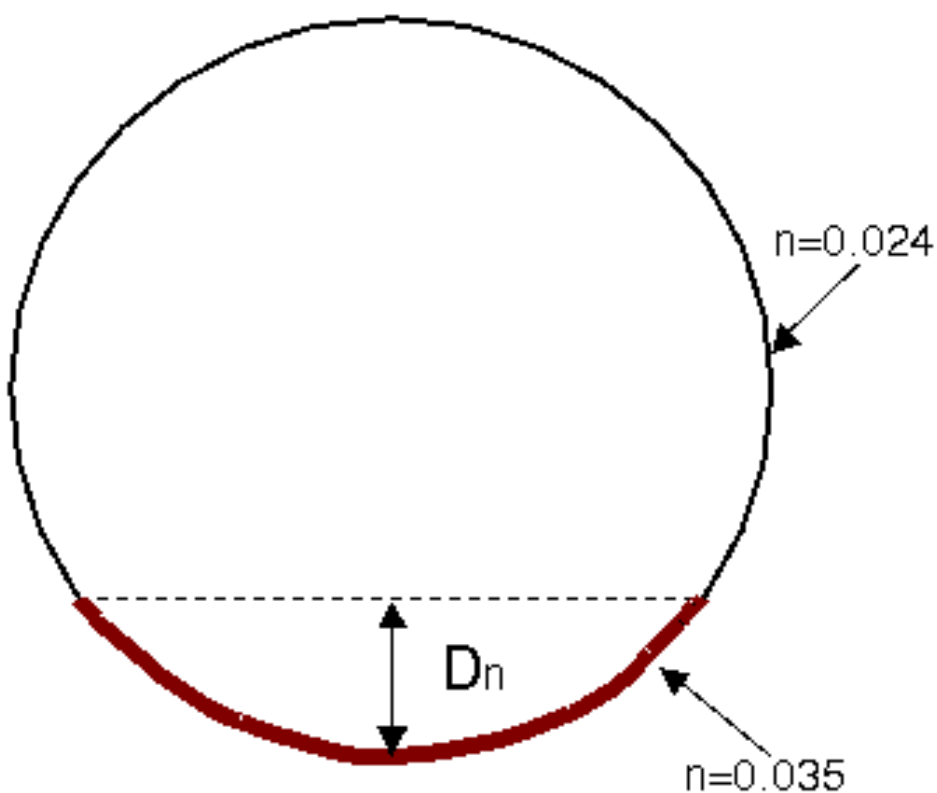
11 SUPERCRITICAL AND MIXED FLOW REGIME INSIDE OF CULVERT

The culvert routines allow for supercritical and mixed flow regimes inside the culvert barrel. During outlet control computations, the program first makes a subcritical flow pass through the culvert, from downstream to upstream. If the culvert barrel is on a steep slope, the program may default to critical depth inside of the culvert barrel. If this occurs, a supercritical forewater calculation is made from upstream to downstream, starting with the assumption of critical depth at the culvert inlet. During the forewater calculations, the program is continually checking the specific force of the flow, and comparing it to the specific force of the flow from the subcritical flow pass. If the specific force of the subcritical flow is larger than the supercritical answer, the program assumes that a hydraulic jump will occur at that location. Otherwise, a supercritical flow profile is calculated all the way through and out of the culvert barrel.

For inlet control, it is assumed that the water surface passes through critical depth near the upstream end of the barrel. The first step is to find the "Vena Contracta" water surface and location. As flow passes through critical depth at the upstream end of the barrel, it goes into the supercritical flow regime. The most constricted depth that is achieved will depend on the barrel entrance shape and the flow rate. The program uses an empirical equation to estimate the location and depth of the Vena Contracta (supercritical) water surface elevation. Once this depth and location are estimated, hydraulic forewater computations are performed to get the water surface profile through the barrel. The program will evaluate and compute a hydraulic jump if either the downstream tailwater is controlling, or if the slope of the barrel is flat enough and long enough that a jump would occur due to barrel control. If a hydraulic jump occurs, and that jump produces a water surface that will fill the barrel, it is assumed that any air pocket in the barrel would burp out and that outflow control will ultimately dictate the upstream energy and water surface elevation.

12 MULTIPLE MANNING'S n VALUES INSIDE OF CULVERT

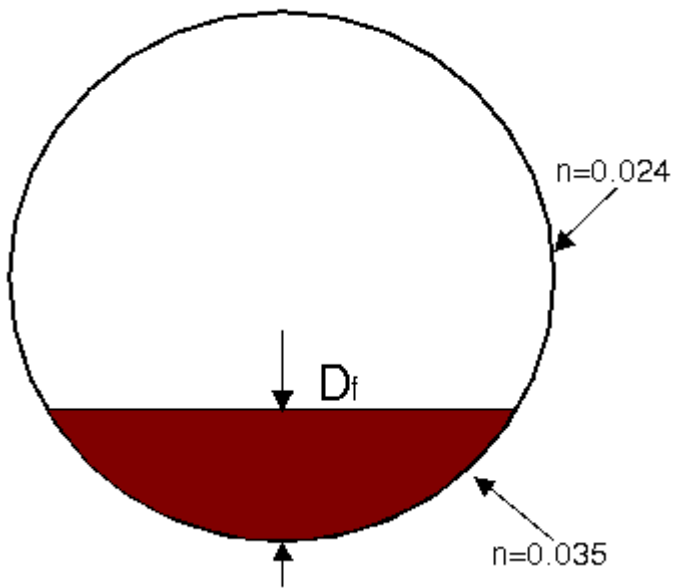
This version of HEC-RAS allows the user to enter two Manning's n values inside of the culvert, one for the top and sides, and a second for the culvert bottom. The user defines the depth inside the culvert to which the bottom n value is applied. This feature can be used to simulate culverts that have a natural stream bottom, or a culvert that has the bottom portion rougher than the top, or if something has been placed in the bottom of the culvert for fish passage. An example of this is shown in the figure below.



When multiple Manning's n values are applied to a culvert, the computational program will use the bottom n value until the water surface goes above the specified bottom n value. When the water surface goes above the bottom n value depth the program calculates a composite n value for the culvert as a whole. This composite n value is based on an equation from Chow's book on Open Channel Hydraulics (Chow, 1959) and is the same equation we use for computing a composite n value in open channel flow (see [\(Composite Manning's \$n\$ for the Main Channel:1\)](#) of this manual).

13 PARTIALLY FILLED OR BURIED CULVERTS

This version of HEC-RAS allows the user to fill in a portion of the culvert from the bottom. This option can be applied to any of the culvert shapes. The user is only required to specify the depth to which the culvert bottom is filled in. An example of this is shown in the figure below. The user can also specify a different Manning's n value for the blocked portion of the culvert (the bottom), versus the remainder of the culvert. The user must specify the depth to apply the bottom n value as being equal to the depth of the filled portion of the culvert.



14 COMPARISON TO THE USGS CULVERT PROCEDURES

Many people have asked how the HEC-RAS culvert routines compare to the USGS culvert procedures outlined in the publication "Measurement of Peak Discharge at Culverts by Indirect Methods" (Bodhaine, 1978), and if the HEC-RAS software would give the same or similar answers to their culvert analysis program (CAP). To prove that HEC-RAS could handle all 6 flow types outlined in the USGS publication, we put models together to replicate 8 of the example data sets in the back of the USGS publication mentioned above. HEC-RAS was able to compute similar upstream water surface elevations, as reported in the USGS publication, for all of the example data sets except number 8. However, we believe the reported result for example problem number 8 is questionable. This will be explained in more detail below. The 8 examples were put together in HEC-RAS by entering the culvert geometry, and all other properties provided in the publication.

Here is a table of the HEC-RAS and USGS answers for the 8 example problems:

Example Number	Flow Rate Q (cfs)	USGS Water Surface (ft)	HEC-RAS Water Surface (ft)	HEC-RAS Flow Classification	USGS Flow Classification
1	729	12.00	11.89	Inlet	1
2	530	10.00	10.68	Inlet	1
3	268	6.00	6.00	Outlet	2
4	523	8.19	8.88	Outlet	2
5	251	6.00	5.94	Outlet	3
6	125	7.00	7.17	Outlet	4
7	120	8.00	8.14	Inlet	5
8*	209	8.00	11.00	Inlet	6

*Note: We think the answer shown in the USGS publication, for example number 8, is questionable. Here is why:

Example 8 is for a circular concrete culvert that is 4.0 feet in diameter. The Manning's n is 0.012, and the culvert has a beveled entrance. The resulting flow rate computed in the example is 209 cfs. The culvert invert is set at an elevation of 1.0 ft at the upstream end, and the top of the culvert is at 5.0 feet inside elevation at the upstream end.

The Culvert Area is $A = 12.5664$ sq. ft.

Therefore $V = Q/A = 209/12.5664 = 16.63$ ft/s inside the culvert at the upstream end.

The velocity head is $V^2/2g = (16.63)^2 / (2 \times 32.3) = 4.3$ feet of velocity head.

Therefore, the energy at the upstream inside end of the barrel must be at least $5.0 + 4.3 = 9.3$ feet of energy head. The energy upstream, outside of the barrel, will be this energy plus an entrance losses to get the flow into the barrel, plus friction losses. Therefore the upstream energy will be greater than 9.3 feet. HEC-RAS computed an upstream energy of 11.0 feet, which we believe is more correct than the 8.00 feet reported in example 8 in the USGS report.

The USGS results of 8.0 feet is based on the assumption of this culvert acting as a Syphon for this particular flow rate. The HEC-RAS computations did not fill the barrel, so we do not think the culvert will act as a Syphon. If the culvert were to act like a Syphon, then the answer would be closer to the USGS culvert routines answer.

Additionally HEC-RAS was able to reproduce all six of the USGS flow classification types. For Type 6, the example 8 problem did not flow as a full barrel for HEC-RAS. However, we took the same data set and lowered the upstream invert to an elevation of 0.0 feet, which put the culvert on a horizontal slope. HEC-RAS did compute that the flow was following the USGS Type 6 classification for this culvert.

We have therefore concluded that HEC-RAS can handle all 6 of the USGS Culvert flow classifications, and can reproduce the results of the CAP program within a reasonable tolerance (Except for example 8, which is in not been resolved at this time). It is also believed that most of the differences in the results are due to the fact they the two programs use different: empirical coefficients entrance losses; friction slope computations (therefore different friction losses); and exit losses.

15 CULVERT FLOW INTO 2D FLOW AREAS

In HEC-RAS 6.7, the momentum from culverts into 2D Flow Areas is included. The culvert outlet momentum is only applicable to the Shallow Water Equation solvers SWE-ELM and SWE-EM. Culvert outflows can have the effect of accelerating or deaccelerating the flow depending on the velocity of the outflow and the local flow velocities. Culvert inflows have no effect on the momentum of 2D flow area cells. The momentum contribution from culvert outflows at a computational face can be written as

$$11) \quad \mathcal{F}_f^- = \mathcal{F}_{e,f}^- + \mathcal{F}_{i,f}^- u_{N,f}$$

where

$$12) \quad \mathcal{F}_{e,f}^- = \frac{1}{\bar{h}_f} \sum_{c \in (L,R)} \frac{\alpha_f^c}{A_c} Q_c^- (\xi_{p,c} \mathbf{V}_c \cdot \mathbf{n}_f)$$

$$\mathcal{F}_{i,f}^- = -\frac{1}{\bar{h}_f} \sum_{c \in (L,R)} \frac{\alpha_f^c}{A_c} Q_c^-$$

and

Q_c^- : culvert outflow at cell c [L^3/T]

$\mathbf{V}_c$: culvert outflow velocity at cell c [L/T]

$\xi_{p,c}$: culvert plunging flow factor at cell c [-]

A_c : cell horizontal area [L^2]

$\bar{h}_f = \alpha_f^L h_L + \alpha_f^R h_R$ [L]

h_c : cell hydraulic depth [L]

$\alpha_f^L = \frac{\Delta x_f^L}{\Delta x_f^L + \Delta x_f^R}$ [-]

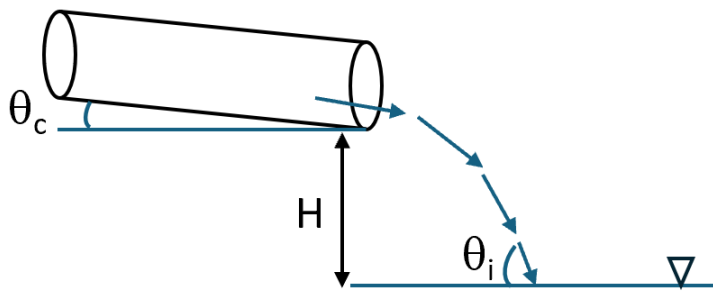
$\alpha_f^R = 1 - \alpha_f^L$ [-]

$\mathbf{n}_f$: unit vector at face f [-]

The calculation of the plunging flow factor is described in the section below.

Plunging Flow Correction

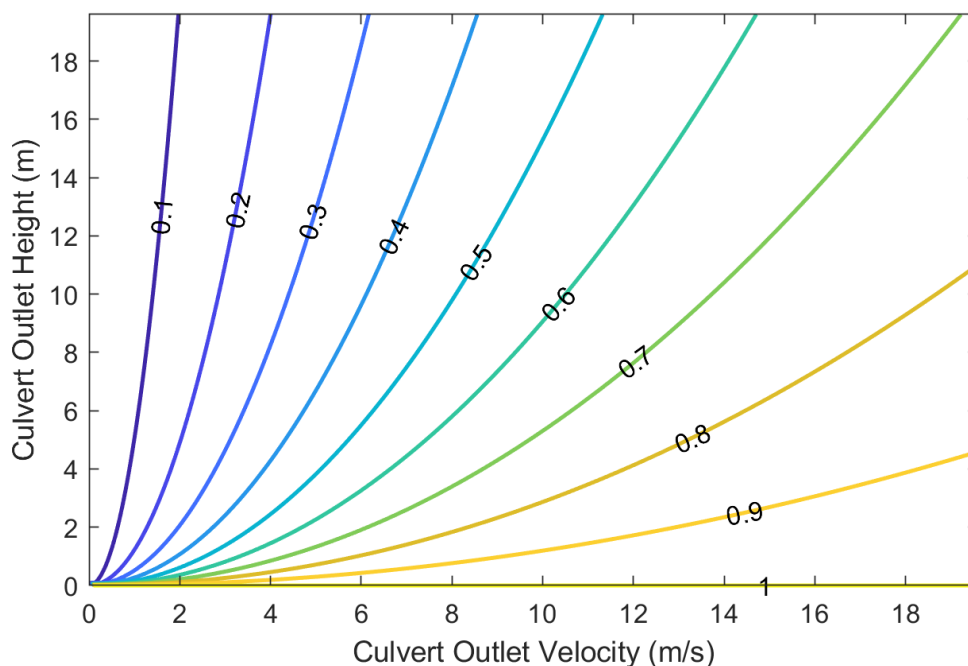
When a culvert outlet is above the local water surface, the culvert flow is considered to be plunging (see figure below).



When plunging flow occurs, only a fraction of the culvert momentum is transferred to the 2D flow model. In HEC-RAS, a simple conceptual model is used to compute the correction factor or plunging flow factor as a function of the impact angle of the culvert flow with the local water surface as

$$\xi_p = \cos(\theta_I)$$

where θ_I is the culvert flow impact angle. The impact angle is computed from projectile theory assuming a horizontal culvert and given the initial horizontal velocity and height of the culvert above the water surface. The figure below shows the culvert plunging factor as a function of the culvert height above the water surface and culvert outlet velocity.



Culvert Plunging Factor as a function of the outlet height and velocity.